Wyndair Daryl Bartolome 40310244 Sprint 2 AI Logs

## Task: Single Job Details API Implementation (Task 6.7)

**Purpose of AI Use:**
Backend API implementation, code generation, testing, and troubleshooting.

**Chat Link or Prompt/Response:**
I'm working on Task 6.7 for our SOEN 341 CareerConnect project. Help me implement a backend endpoint (GET /api/jobs/:id) using Express and SQLite to retrieve individual job postings, including their descriptions and requirements. The API should validate job IDs, return appropriate HTTP status codes, and only retrieve open job postings. Guide me through the implementation step by step while preserving the existing backend code. Also, help me create automated tests using Jest and Supertest to verify the endpoint and ensure existing functionality is not affected.

**AI-Suggested Content:**
ChatGPT provided guidance for implementing the GET /api/jobs/:id endpoint to retrieve individual job postings from the SQLite database. It suggested input validation, handling nonexistent or closed jobs, and returning appropriate HTTP status codes. It also generated Jest and Supertest cases to verify the API's functionality.

**Validation:**
The implementation was reviewed against the existing backend structure and task requirements. All 7 new API tests passed, including valid, invalid, nonexistent, and closed job scenarios. The complete backend test suite was also executed, with 36 out of 36 tests passing.

**Decision:**
**Accepted after testing and validation.**

**Reflection:**
The AI helped clarify how Express routes interact with the SQLite database and how to handle API responses. The automated tests were useful for verifying the endpoint without affecting existing backend functionality.

**Responsible Person:**
Wyndair Daryl Bartolome

# Task: Application API Security and Regression Tests (Task 10.6)

**Purpose of AI Use:**
Automated test generation, security validation, and regression testing.

**Chat Link or Prompt/Response:**
I'm working on Task 10.6 for our SOEN 341 CareerConnect project. Help me create automated backend security and regression tests for the existing job application API (POST /api/applications) using Jest and Supertest. The tests should cover successful submissions, duplicate applications, authentication and authorization, invalid or nonexistent job IDs, closed job postings, resume ownership, and application data privacy. Please review the existing backend structure before suggesting code, avoid modifying existing API functionality unnecessarily, and explain how to execute and validate the tests.

**AI-Suggested Content:**
ChatGPT generated Jest and Supertest cases for the existing POST /api/applications endpoint. The tests covered successful submissions, duplicate applications, authentication and authorization restrictions, invalid job IDs, closed postings, resume ownership, and application data privacy.

**Validation:**
The generated tests were reviewed against the existing application API and database structure. The test file was added to the backend, and all 17 automated tests passed using npm test -- applications.test.js --ci. No changes to the existing application API were required.

**Decision:**
**Accepted after testing and validation.**

**Reflection:**
The AI helped identify important security and error-handling scenarios that should be tested. It also demonstrated how Jest and Supertest can verify API behavior using a temporary test database without modifying actual application records.

**Responsible Person:**
Wyndair Daryl Bartolome